

Ex. No: 7 Use AFLogical OSE to extract data from an Android

Date:

Description:

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

1. Download AFLogical OSE:

- Visit the AFLogical OSE GitHub page and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AFLogical OSE requires Java Make sure you have Java installed on your machine. You can download it from the official Java website.

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the Android Developer's website and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4.Enable USB Debugging on the Android Device:

- On the Android device, go to Settings About Phone and tap Build Number times to enable Developer Options.
- Go to Settings Developer Options and enable USB Debugging

Step 2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:
adb devices

Download AFLogical OSE:

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step 3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE: Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogical OSE APK on the device: `bash adb install aflogical.apk`
- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogical OSE App on the Device:

- You'll see options to extract various types of data (eg, contacts, SMS, MMS, call logs). Select the data types you to extract
- On the Android device, open the AFLogical app.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in files on the device's storage, typically in a directory named aflogical.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer.
Run:
`bash adb pull/sdcard/aflogical/path/to/destination`
- Replace/path/to/destination with the directory path where you want to save the extracted data on your computer.

1. Verify the Data:

Navigate to the destination directory and check the esv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running.

Bash adb

uninstallcom.viaforensics.android.aflogical

2. Disconnect the Device:

- Safely disconnect the Android device from your computer. By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

Output:

Installing JAVA:

```
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\siddu>java -version
java version "26.0.2.1" 2026-08-18
Java(TM) SE Runtime Environment (build 26.0.2.1+1-7)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.2.1+1-7, mixed mode, sharing)

C:\Users\siddu>
```

Installing ADB Tools:

```
C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>
```

ADB device connection:

```
C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
ZD2227X3N4      device

C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>
```

Installing aflogical.apk:

```
C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>adb install
com.viaforensics.android.aflogical_ose_1.5.2_QSE_152.apk
Performing Streamed Install
Success

C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>
```

Capturing SMS and Call logs:

```
C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>.\adb.exe shell content query --uri
content://sms > sms.txt

C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>.\adb.exe shell content query --uri
content://call_log/calls > call_logs.txt

C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>
```

Extracted SMS logs:

```
Row: 0 _id=6806, thread_id=2294, address=9f1982675638, person=NULL, date=17887599138655, date_sent=17887599138658, protocol=0, read=0, status=1, type=1, reply_path_present=0,
subject=proto(cjkzWb55nb28n86dBUWskcz8pXCSwHdLl1lc2bW22uLy45FCIAkh0SwA8s3QNR7p1zn57E6Pk, body=Dear Customer, You have a missed call from +9194932675638. The last missed call was at
16:58 AM on 07-Sep-2026. Thankyou, Team Jio, service_center=+919044900412, locked=0, sub_id=2, error_code=0, creator=com.google.android.apps.messaging, seen=1
Row: 1 _id=6807, thread_id=1887, address=3K1HSE5W5-S, person=NULL, date=1788844789072, date_sent=1788844785600, protocol=0, read=0, status=1, type=1, reply_path_present=0,
subject=proto(cjkzWb55nb28n86dBUWskcz8pXCSwHdLl1lc2bW22uLy45FCIAkh0SwA8s3QNR7p1zn57E6Pk, body=CROMAINVESTECORHRRIVATEJITED, on 29-08-2026 reported your F&Kmail, Rs.8,000 0
securities balwith The broker-NSE, service_center=+917388307506, locked=0, sub_id=2, error_code=0, creator=com.google.android.apps.messaging, seen=1
Row: 2 _id=6808, thread_id=2237, address=3X-TSRSCM-S, person=NULL, date=1788533923394, date_sent=1788533919800, protocol=0, read=0, status=1, type=1, reply_path_present=0,
subject=proto(cjkzWb55nb28n86dBUWskcz8pXCSwHdLl1lc2bW22uLy45FCIAkh0SwA8s3QNR7p1zn57E6Pk, body=RTR4444428174
Tn:422715
Dt:05-09-26 Dep.Time=09:47 Hrs.
Frm KRNT to NDU
Cls:SLL
P1-94,34
P2-54,29
P3-54,32
Boarding allowed from KRNT only
Chart Prepared
Url for coach position: https://enquiry.indianrail.gov.in/wntes/c/wag6fjNwa9d7WeJongKd81
Download RailOne for latest updates https://railone.indianrailways.gov.in
For Enquiry/Complaint/Assistance, please dial 139 FR-CRIS, service_center=+91701978801, locked=0, sub_id=2, error_code=0, creator=com.google.android.apps.messaging, seen=1
Row: 3 _id=6807, thread_id=3771, address=3K1HSE5W5-S, person=NULL, date=178853867713, date_sent=178853635000, protocol=0, read=0, status=1, type=1, reply_path_present=0,
subject=proto(cjkzWb55nb28n86dBUWskcz8pXCSwHdLl1lc2bW22uLy45FCIAkh0SwA8s3QNR7p1zn57E6Pk, body=Bearer white details based on unsolicited trip through
Whatsapp, telegram,SMS,calls,YouTube,etc& take an informed decision. Visit https://bit.ly/3Q0XSko -NSE, service_center=+9170204705016, locked=0, sub_id=2, error_code=-1,
creator=com.google.android.apps.messaging, seen=1
Row: 4 _id=6806, thread_id=1886, address=3J1Bf4Y-S, person=NULL, date=1788532742702, date_sent=1788532691000, protocol=0, read=0, status=1, type=1, reply_path_present=0,
subject=proto(cjkzWb55nb28n86dBUWskcz8pXCSwHdLl1lc2bW22uLy45FCIAkh0SwA8s3QNR7p1zn57E6Pk, body="=0";
Auto 04-Sep-26 17:21 ವಯಸ್ವಿರ್ಮಿ ರೆಡ್‌ಲೈಟ್ ರೆಡ್‌ಲೈಟ್ 100% ನಿರ್ದಿಷ್ಟಗೊಂಡು.
ವಿಟೆ ನಂಬರ್: 8874999510
ರೇವ್‌ನಾರಿ ರೇಡ್‌ಲೈಟ್ 2 GB
ರೇವ್‌ನಾರಿ ರೇಡ್‌ಲೈಟ್ 2GB: 09:27 ನವೆಂಬರ್ ಅಂದಾಜುಬೇರೆ ಕೂಡುವುದು.
ಮಿ ನವೆಂಬರ್ ಅಂದಾಜು ರೇಡ್‌ಲೈಟ್ ರೇಡ್‌ಲೈಟ್, ಮಿ ವಾಟರ್‌ನಿಂದ ವರ್ಗ 66 kbps ವರೆಗೆ ಕಳುಹಿಸುವುದು.
MyJio ಯಾವ ದಾರಿ ರೇಡ್‌ಲೈಟ್ ರೇಡ್‌ಲೈಟ್ ವೇದಾಂತಕ್ಕೆ ಕೊಡುವುದು: http://tiny.jio.com/dhydatapack
```

Extracted Call logs:

```
Row: 0 formatted_number=NULL, duration=50, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number, countryiso=IN, cnap_name=, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0,
cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1,
numbertype=NULL, features=68, transcription=NULL, last_modified=1729788812435, _id=8385, new=0, date=1718892698184, name=Seenu, type=1, presentation=1,
via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL,
matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1,
Row: 1 formatted_number=NULL, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=NULL, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number, countryiso=IN, cnap_name=NULL, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0,
cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1,
numbertype=NULL, features=1, transcription=NULL, last_modified=1729788812435, _id=8386, new=0, date=1718892698184, name=Seenu, type=1, presentation=1,
via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL,
matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1,
Row: 2 formatted_number=NULL, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number, countryiso=IN, cnap_name=NULL, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0,
cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1,
numbertype=NULL, features=1, transcription=NULL, last_modified=1729788812435, _id=8386, new=0, date=1718892698184, name=Seenu, type=1, presentation=1, via_number=,
numberlabel=Mobile, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL,
matched_number=NULL, transcription_state=0, data_usage=NULL, video_call=0,
Row: 3 formatted_number=+91 77991 68016, duration=38, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0,
post_dial_digits=, call_screening_app_name=NULL, priority=0, number=+917799168016, countryiso=IN, cnap_name=+917799168016, photo_uri=, geocoded_location=India,
missed_reason=0, block_reason=0, cached_convert_number=NULL,
subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=0, features=64,
transcription=NULL, last_modified=1729788812435, _id=8499, new=0, date=1720318990704, name=, type=1, presentation=1, via_number=, numberlabel=Mobile,
normalized_number=+917799168016,
```

Extracted Device Properties:

```
PS C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Device Model:" (.\adb.exe shell
getprop ro.product.model)
Device Model: moto g62 5G
PS C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Android Version:" (.\adb.exe
shell getprop ro.build.version.release)
Android Version: 13
PS C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Security Patch Level:" (.
.\adb.exe shell getprop ro.build.version.security_patch)
Security Patch Level: 2025-07-01
PS C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>
```

Uninstalling aflogical.apk:

```
C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-  
tools>.\adb.exe uninstall --user 0  
com.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk  
C:\Users\siddu\Downloads\platform-tools-latest-windows\platform-tools>
```

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The experiment successfully performed a logical extraction of data from the Android device using the AF Logical OSE framework via ADB interface connection. The tool bypassed file system access limitations to cleanly pull device properties, SMS logs, and call records into structured.csv files, which were verified on the local host before safely uninstalling the APK tool.

